

Challenge (Habanero)

- Q1.** Solve for x : $2^x = 32$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$
- Q2.** A gamer's score doubles every level. If the score is 2^5 at level 5, what is the score at level 7?
(A) 2^5
(B) 2^6
(C) 2^7
(D) 2^8
(E) 2^9
- Q3.** If $3^x = 81$, then $x =$
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7
- Q4.** Solve for y : $4^y = 256$
(A) $y = 2$
(B) $y = 3$
(C) $y = 4$
(D) $y = 5$
(E) $y = 6$
- Q5.** A coding error causes a program to use 2^x times more data. If $x = 3$, how many times more data is used?
(A) 2 times
(B) 4 times
(C) 6 times
(D) 8 times
(E) 10 times
- Q6.** If $5^x = 125$, then $x =$
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** Solve for z : $2^z = 16$
(A) $z = 2$
(B) $z = 3$
(C) $z = 4$
(D) $z = 5$
(E) $z = 6$
- Q8.** A scoreboard's points are calculated by 3^x . If $x = 2$, how many points are scored?
(A) 6 points
(B) 8 points
(C) 9 points
(D) 12 points
(E) 15 points

Open Ended Questions (FRQ)

Q9. Solve for x : $x^2 + 2^x = 10$

Q10. A gamer's score increases by 2^x each level. If the score at level 3 is 12, find the score at level 5.